AI-Powered Cricket Commentary

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Objective

Primary Goal

Generate human-like cricket commentary from highlight videos.

Sub Goals

- **Event Recognition:** Detect meaningful events (e.g., deliveries, boundaries) from video frames.
- **Commentary Generation:** Generate concise, vivid cricket commentary using a vision-language model (GPT-4o OR GPT-4o-mini).
- **Audio Synthesis:** Convert generated commentary text into realistic speech audio using gTTS.

Why it Matters?

- Enable **automated sports broadcasting**.
- Improve **fan engagement** and **content generation.**
- Support real-time and multilingual commentary in future applications.

Workflow

Part 1

Video Input → Frame Extraction → Event Detection

Part 2

GPT-4o Commentary Generation → Text Commentary

Part 3

Google Text-to-Speech Conversion → Audio Commentary

Part 1 - Event Recognition

1. **Input:** 30–40 second cricket highlight video (.mp4 or .mov).
2. **Frame Extraction:**
   - Read video using OpenCV.
3. **Frame Difference Scoring:**
   - Convert each frame to grayscale.
   - Compute absolute pixel-wise difference between consecutive frames.
   - Calculate a "motion score" for each frame.
4. **Top-K Frame Selection:**
   - Sort frames based on motion scores.
   - Select top K frames representing the highest visual changes (likely key events like ball deliveries, shots, celebrations).
5. **Minimum Gap Filtering:**
   - Enforce a 3-second minimum time gap between selected frames to avoid over-segmentation or repeated detections.
6. **Segmentation:**
   - Create small clips or isolate events based on selected frames.

Part 2 - Commentary Generation

1. **Input:** Selected event frames (base64 encoded).
2. **Frame Sampling Optimization:**
   - Sample every 25th/30th/40th frame to reduce input size while maintaining event context.
3. **Prompt Engineering:**
   - Design two types of prompts:
     - **Single Event Prompt:** "Describe this delivery as a cricket commentator."
     - **Multiple Event Prompt:** "This video has multiple deliveries. Number each delivery's commentary separately."
   - Instruct GPT to avoid mentioning: Player names, Crowd details, Team colors
   - Focus only on cricket action (delivery, shot, outcome).
4. **Receive Model Output:** Model generates vivid cricket commentary text per event.

Part 2 - Ablation Test

1. **Goal:** Find the minimum number of frames needed for accurate commentary generation.
2. **Single Event:** Every 100th frame was sufficient for GPT-4o and GPT-4o-mini.
3. **Multiple Events:** GPT-4o required denser sampling (every 30th–50th frame) for accurate narration.
4. **Trade-Off:** Sparse sampling reduces cost without major quality loss for single events.
5. **Limitation:** In subtle/difficult plays, commentary was fluent but sometimes inaccurate.

Part 3 - Text-to-Speech Conversion

1. **Input:** Commentary text generated from GPT-4o.
2. **gTTS Synthesis:**
   - Use gTTS Python library to send text to Google's TTS cloud API.
   - Specify the target language (e.g., 'en' for English).
   - No need for GPU or local model weights — synthesis happens server-side.
3. **Audio File Generation:**
   - Receive MP3 audio output for each commentary line.
   - Save each file locally (e.g., commentary_1.mp3, commentary_2.mp3,..)
   - The modular audio files allow flexible playback and editing.
4. **Batch Processing:**
   - If multiple events are detected in the video, process them sequentially.
   - Generate and save separate .mp3 files for each event commentary.
   - Organize files for easy mapping back to the corresponding video events.
5. **Playback & Export:**
   - Play the audio files directly using standard media players or programmatic methods (e.g., IPython.display in Jupyter).

Experiments and Learnings

Part 1

Output:
- List of key event frames or short segments corresponding to cricket deliveries, boundaries, or dismissals.

Part 2

Output:
- Short, event-specific text commentary for each cricket delivery.
- Example:
  - "The batsman drives elegantly through covers for a boundary."
  - "Massive six over long-on! A clean strike."

Part 3

Output:
- Clear MP3 audio files of commentary.
- Example: "commentary_event1.mp3", "commentary_event2.mp3", ready for voiceover integration.
- In future extensions, the audio clips can be merged with video highlights to produce fully narrated cricket highlight reels.

Conclusion

1. Successfully built core components: event detection, GPT-4o-based commentary generation, and gTTS-based audio synthesis.
2. Event detection reliably segmented cricket highlights into meaningful individual events.
3. GPT-4o models generated fluent and contextually accurate cricket commentary from video frames.
4. gTTS provided quick and efficient audio synthesis for initial testing.
5. Ablation study optimized frame sampling strategy, balancing quality and API efficiency.
6. Identified model limitations in handling highly ambiguous or visually subtle events.

Future Work

1. **Next Steps:**
   - Improve automatic event detection.
   - Upgrade TTS model for professional-grade voiceovers.
   - Merge audio back onto video highlights.

2. **Vision:** Fully automated cricket commentary generation system.

Demo

Thank You!